Fangda Ye

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Research Interests

I aim to build **general-purpose agents** that take on open-ended tasks **from the digital to the physical world**, run continuously over long horizons, and keep improving from their own experience. My work so far builds and evaluates deep research agents (multimodal search, grounded long-form generation, agentic post-training, process-level evaluation), and extends to long-term memory for proactive assistants, a self-improving production agent framework, and test-time training for robot memory. The settings I care about are **personal assistants**, **long-horizon agents** and **agents for physical AI**, built on **agent training**, **self-improvement**, and **context and memory management**.

Education

National University of Singapore (NUS) — Singapore
Master of Computing (Computer Science) · GPA: 4.88 / 5.00 — *Aug 2024 – Jun 2026*

- Academic Excellence Award (AY2025/26); Dean's List in both semesters of AY2024/25.
- Thesis: *Proactive Demand Detection in Multi-Turn Conversations: Benchmark, Method, and Evaluation*. Advisor: Prof. Shuicheng Yan.

Central South University — Changsha, China
Bachelor of Engineering · GPA: 90.78 / 100 — *Sep 2019 – Jun 2023*

Publications

* equal contribution

Conference papers

[C1] **MiroEval: Benchmarking Multimodal Deep Research Agents in Process and Outcome**
Fangda Ye*, Yuxin Hu*, Pengxiang Zhu*, Yibo Li*, Ziqi Jin, …, Bryan Hooi, Bo An, Shuicheng Yan, Lidong Bing
Advances in Neural Information Processing Systems (NeurIPS), Datasets & Benchmarks Track, 2026 [arXiv] [Code]

[C2] **Deep-Reporter: Deep Research for Grounded Multimodal Long-Form Generation**
Fangda Ye*, Kuicai Dong*, Zhifei Xie*, Yuxin Hu*, Yihang Yin, Shurui Huang, Shikai Dong, Chen Zhang, Jianzhu Bao, Shuicheng Yan
Annual Meeting of the Association for Computational Linguistics (ACL), Main Conference, 2026 [Paper] [arXiv]

[C3] **Seeing the Unseen: Unified Visible–Invisible Motion for Physically Consistent Video Generation**
Mingyang Bao, **Fangda Ye**, Xihao Chen, Bobo Li, Xiangtai Li, Shengqiong Wu
Advances in Neural Information Processing Systems (NeurIPS), 2026

[C4] **Doc-Researcher: A Unified System for Multimodal Document Parsing and Deep Research**
Kuicai Dong*, Shurui Huang*, **Fangda Ye***, Wei Han, Zhi Zhang, Dexun Li, Wenjun Li, Qu Yang, Gang Wang, Yichao Wang, Chen Zhang, Yong Liu
The ACM Web Conference (WWW), 2026 [arXiv]

Under review

[U1] **Temporal Contrastive Test-Time Training for Robotic Manipulation**
Yihang Yin*, **Fangda Ye***, Shiguang Wu, Kunhao Pan, Ruiyang Xu, Ruyi Gan, Hao Wang
Under review

Technical reports

[T1] **MiroThinker-1.7 & H1: Towards Heavy-Duty Research Agents via Verification**
MiroMind Team (contributor; **Fangda Ye** among 41 authors)
Technical report, 2026 [arXiv]

[T2] **PASK: Toward Intent-Aware Proactive Agents with Long-Term Memory**
Zhifei Xie, Zongzheng Hu, **Fangda Ye**, Xin Zhang, Haobo Chai, …, Deheng Ye, Chunyan Miao, Shuicheng Yan
Technical report, 2026 [arXiv]

Research Experience

MiroMind (now Apodex) — Dec 2025 – Jun 2026
AI Scientist Intern · Lead: Lidong Bing · Mentors: Yue Deng, Lei Wang, Bin Wang — *Singapore*

- **Evaluation.** Co-first author of **MiroEval** (NeurIPS 2026 Datasets & Benchmarks): 100 real-world tasks (70 text, 30 multimodal) that score deep research agents on report synthesis, factuality, and an audit of the research process itself. Found that process quality predicts the final outcome. MiroEval also served as the internal evaluation suite for **MiroThinker-1.7 & H1**.

– **Agentic post-training for wide search.** Post-trained Qwen3.5-35B-A3B to issue and combine searches in parallel (+35% relative on WideSearch-Bench), owning the data pipeline end to end: 10K candidate tasks from 11 APIs and 8 query patterns, filtered to 2K hard seeds by self-consistency and two-model quality checks.
– **Agent framework.** Designed memory management, context management, and retry and error recovery for the MiroFlow agent framework; shipped in MiroFlow v1.7, which reached 82.7 on GAIA and 75.3 on BrowseComp-ZH.

Learning and Vision (LV) Lab, National University of Singapore Aug 2025 – May 2026
Graduate Researcher, with Prof. Shuicheng Yan *Singapore*

– **Proactive demand detection.** Master's thesis within the **PASK** project: deciding, turn by turn, whether an assistant listening to a conversation should step in without being asked.
– **Real-world benchmark.** Built **LatentNeeds-Bench**: 100 real multi-turn conversations (3,936 turns) from podcast transcripts, annotated per turn with a two-level taxonomy of 11 demand types. Nine frontier LLMs reach a median of only 66.2% balanced accuracy with prompting.
– **Post-training with RL.** Trained **InnerMind**, a Qwen3-30B-A3B detector, with SFT and then GRPO on a rule-plus-judge reward, behind a context engine for memory and prompt assembly. Reaches 84.2% balanced accuracy at 1.4 s latency, above Gemini-3-Flash (80.8%) and 25.3 points over the base model.

Huawei Singapore Research Center (Noah's Ark Lab) May 2025 – Dec 2025
AI Researcher Intern, with Dr. Kuicai Dong *Singapore*

– **Multimodal report generation.** Built **Deep-Reporter** end to end (co-first author, ACL 2026): an agent that writes long research reports with images, grounded in what it retrieves. It plans each report as a checklist of sections, searches text and images in separate streams with relevance filtering, and writes section by section with recurrent context management.
– **Agentic SFT.** Fine-tuned Qwen3-32B on 8K curated agentic traces, raising its overall score from 27.2 to 37.9 (naive RAG: 5.2; STORM-style agent: 15.6) on **M2LongBench**, built for this work: 247 tasks in 9 domains over a fixed sandbox of 95K images and 108M text chunks.
– **Multi-agent system.** For **Doc-Researcher** (co-first author, WWW 2026), deep research over long multimodal documents, designed the multi-agent workflow: a Planner agent that narrows the document pool by 60–80% and picks the retrieval granularity, and Searcher and Refiner agents that loop until the evidence suffices. Removing the planner costs 5.7–8.3 points; the full system reaches 50.6%, 3.4× the best baseline.

Industry Experience

TikTok Jun 2026 – Present
AI Engineer, ranking models for e-commerce short videos in the main feed *Singapore*

– **Self-improving agent framework.** Built a recursive self-improvement (RSI) framework that makes coding agents more capable and more autonomous with use: each task is distilled into a shared, structured memory of domain facts, tools, workflows and failure diagnoses, so everyone's work improves it. Adopted across the department.
– **Concurrent context management.** Tasks evolve the shared memory in parallel, each in an isolated workspace; a consolidation agent merges its findings into place by explicit rules, keeping memory consistent instead of piling up conflicting notes. Raw sessions stay indexed as a last resort; a capability registry stops long-running agents forgetting their tools; tiered guardrails (autonomous, confirm first, never) keep production actions safe.
– **Applied across the stack.** Used for feature engineering, model development, the online serving pipeline, training infra and incident diagnosis; added 100+ features in two weeks (previously 0.5–1 week each).
– **Training acceleration.** Speeding up ranking-model training with user-level training (ULT) and system-level optimizations (custom Triton kernels, attention tiling), reaching +40% per-GPU throughput at unchanged AUC.

Honors and Awards

Academic Excellence Award, School of Computing, NUS — Jun 2026
For outstanding academic performance in AY2025/26; one recipient in the programme each year.
Dean's List, School of Computing, NUS (AY2024/25 Semester 1 and Semester 2) — 2025
Outstanding Intern, Huawei Singapore Research Center — 2025
National Second Prize, China Postgraduate Mathematical Contest in Modeling — 2023
Outstanding Graduate and Outstanding Undergraduate Thesis, Central South University — 2023

Academic Service

Reviewer, LCFM Workshop at NeurIPS 2026 and *ENGINEERING Information Technology & Electronic Engineering*

Skills

LLM post-training: SFT and LoRA, RL for LLMs (GRPO, DPO), agentic trajectory curation, multi-GPU training
Agents and evaluation: self-improving agents, context and memory management, multi-agent workflows (LangGraph, MCP), multimodal retrieval (Milvus), LLM-as-judge evaluation, benchmark construction
Recommender Systems: large-scale ranking models, Transformer feature interaction, training acceleration (ULT)
Programming: Python, PyTorch, Triton **Languages:** Chinese (native), English (fluent)